IEOR 221 - HW 4 - Jie Wen

1

Arbitrage exists (put is underpriced).

For a non-dividend stock, the European put must satisfy the lower bound

$$p \geq \max\{Ke^{-rT} - S, 0\}$$

Here ($K = 50$, $S = 47$, $r = 6\%$), ($T = 1/12$) year, so

$$Ke^{-rT} = 50e^{-0.06/12} \approx 49.7506, \qquad Ke^{-rT} - S \approx 2.7506$$

Market put price ($p = 2.50 < 2.7506$) $\Rightarrow$ **violates the lower bound**.

Arbitrage strategy (zero net cost at $t = 0$)

1. **Borrow** ($S + p = 47 + 2.50 = 49.50$) at the risk-free rate for ($T = 1$) month.
2. **Buy 1 share** (pay $S = 47$).
3. **Buy 1 put** (pay $p = 2.50$).

Net cash today: ($+49.50 - 47 - 2.50 = 0$).

Payoff at T

The debt to repay is ($(S + p)e^{rT} = 49.50e^{0.06/12} \approx 49.7481$).

- If ($S_T \leq K$): exercise the put, receive ($K = 50$).
  Profit ($= 50 - 49.7481 \approx 0.2519$).
- If ($S_T > K$): let the put expire, sell the stock for ($S_T \geq 50$).
  Profit ($= S_T - 49.7481 \geq 50 - 49.7481 \approx 0.2519$).

Guaranteed arbitrage profit $\approx$ **$0.252 per share** (about $25.19 per standard 100-share option contract).

2

Using put–call parity with **discrete dividends**:

$$c - p = S_0 - \mathrm{PV}(D) - Ke^{-rT} \quad \Longrightarrow \quad p = c - S_0 + \mathrm{PV}(D) + Ke^{-rT}$$

Given $c = 2$, $S_0 = 29$, $K = 30$, $T = 0.5$ years, $r = 10\%$ (assume continuous comp.), and dividends of $0.50 at $t_1 = 2/12$ and $t_2 = 5/12$ years:

$$\mathrm{PV}(D) = 0.50e^{-0.10 \cdot \frac{2}{12}} + 0.50e^{-0.10 \cdot \frac{5}{12}} \approx 0.97133$$

$$Ke^{-rT} = 30e^{-0.10 \cdot 0.5} \approx 28.53688$$

Therefore,

$$p = 2 - 29 + 0.97133 + 28.53688 \approx 2.508 \approx \mathbf{2.51}$$

3

Construct the portfolio (cost today):

- Long 1 call at $K_1 \Rightarrow$ pay c_1
- Short 2 calls at $K_2 \Rightarrow$ receive $2c_2$
- Long 1 call at $K_3 \Rightarrow$ pay c_3

Net cost = $c_1 - 2c_2 + c_3$.

Payoff at expiry (T):

Let $\Delta \equiv K_2 - K_1 = K_3 - K_2 > 0$.

$$\Pi_T(S_T) = \max(S_T - K_1, 0) - 2\max(S_T - K_2, 0) + \max(S_T - K_3, 0)$$

Compute piecewisely:

- If $S_T \leq K_1$: $\Pi_T = 0$.
- If $K_1 < S_T \leq K_2$: $\Pi_T = (S_T - K_1) \geq 0$.
- If $K_2 < S_T \leq K_3$: $\Pi_T = (K_3 - S_T) \geq 0$.
- If $S_T \geq K_3$: $\Pi_T = 0$.

So $\Pi_T(S_T) \geq 0$ for all S_T, with a "tent" shape that peaks at $S_T = K_2$ with height Δ.

No-arbitrage implication:

A portfolio whose payoff is everywhere nonnegative must have a **nonnegative** initial cost. Hence:

$$c_1 - 2c_2 + c_3 \geq 0 \quad \Longrightarrow \quad c_2 \leq \frac{c_1 + c_3}{2}$$

That is, the European call price as a function of strike is **convex** (for common maturity), and equally spaced strikes give the midpoint inequality above.

4

Initial investment (net premium):

$4.75 - 2.15 = \mathbf{\$2.60}$ (per share; \$260 per standard 100-share contract).

Payoff at expiry (excluding initial investment):

$\text{Payoff} = \max(29 - S_T, 0) - \max(25 - S_T, 0)$

- (a) $S_T = 23$: $(29 - 23) - (25 - 23) = 6 - 2 = \mathbf{\$4.00}$
- (b) $S_T = 28$: $(29 - 28) - \max(25 - 28, 0) = 1 - 0 = \mathbf{\$1.00}$
- (c) $S_T = 33$: $0 - 0 = \mathbf{\$0.00}$

5

Let

- $S \equiv S_{T_1}$ be the stock price at T_1,
- $\tau = T_2 - T_1$ be the time still left on the long call,
- $c_\tau(S; K_2)$ be the **time-T_1 value** of a European call with strike K_2 and time to expiry τ,
- Π_0 be the **net initial premium** paid at $t = 0$ (debit if > 0, credit if < 0).*

Profit at T_1 as a function of S is

$$\Pi_{T_1}(S) = c_\tau(S; K_2) - \max(S - K_1, 0) - \Pi_0$$

Because $c_\tau(\cdot; K_2)$ is increasing and convex in S:

(a) $K_2 > K_1$ (long higher strike, short lower strike)

Piecewise form:

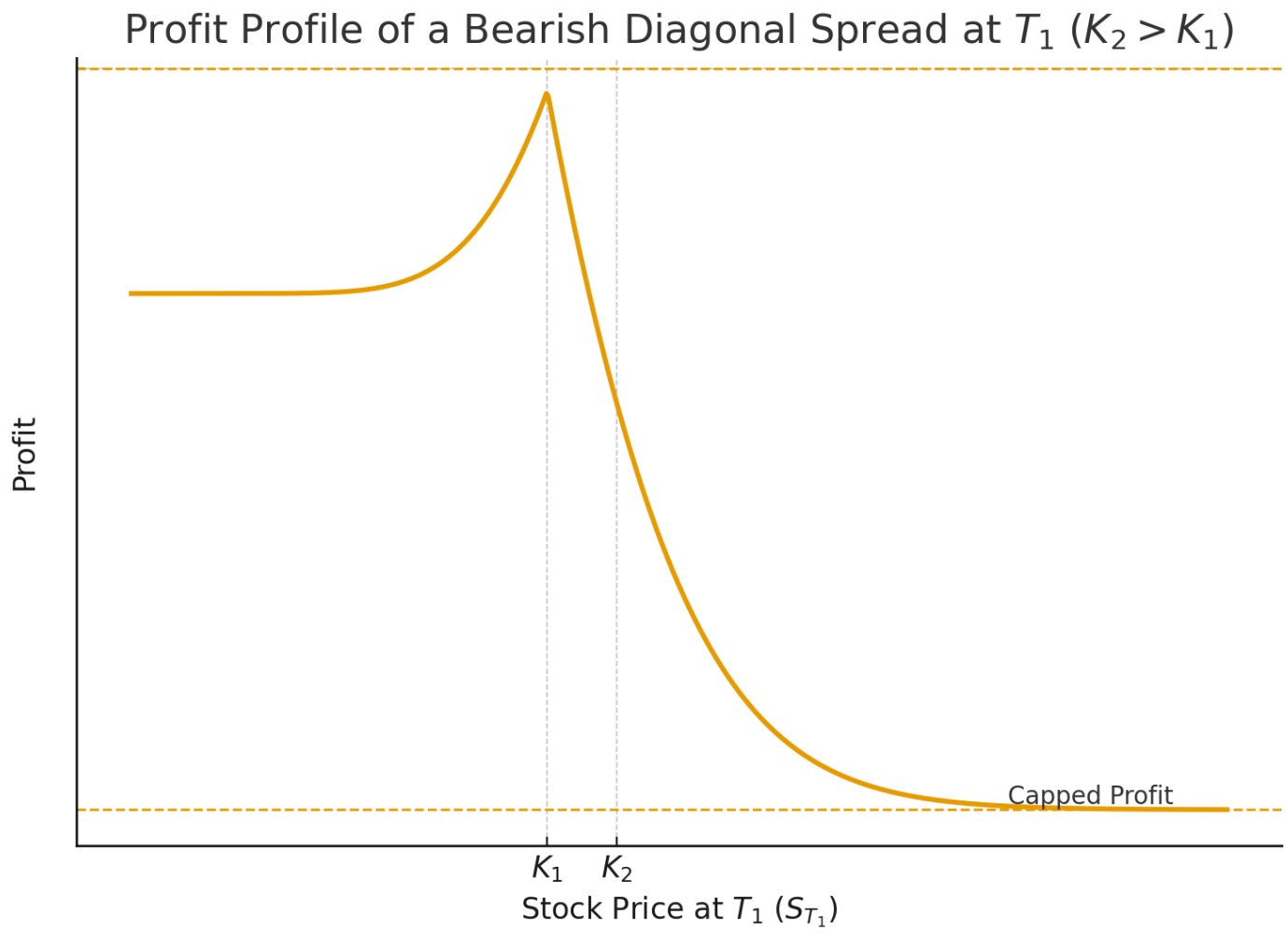
- $S \le K_1$: $\Pi_{T_1}(S) = c_\tau(S; K_2) - \Pi_0$
- $S > K_1$: $\Pi_{T_1}(S) = c_\tau(S; K_2) - (S - K_1) - \Pi_0$

Qualitative shape:

- Left tail (low S): $\approx -\Pi_0$ (the long call is cheap; mainly "eat" the initial debit).
- Around $S \approx K_1$: **hump/peak** (the short call expires, keep its premium; the long call still has time value).
- Right tail (very high S): approaches a **flat** level because

  $$c_\tau(S; K_2) \approx S - K_2 e^{-r\tau} \quad \Rightarrow \quad \Pi_{T_1}(S) \approx K_1 - K_2 e^{-r\tau} - \Pi_0,$$

  typically a **capped** (often negative) plateau.
  Bearish diagonal: best near K_1; upside is capped (can even lose if rally is too big).



(b) $K_2 < K_1$ (long lower strike, short higher strike)

Piecewise form:

- $S \le K_1$: $\quad \Pi_{T_1}(S) = c_\tau(S; K_2) - \Pi_0$
- $S > K_1$: $\quad \Pi_{T_1}(S) = c_\tau(S; K_2) - (S - K_1) - \Pi_0$

Same formula, **different shape** because the long call is now **more in-the-money** for a given S:

- Left tail (low S): $\approx -\Pi_0$ again.
- Around $S \approx K_1$: you still get a local lift from the short decaying away and the long retaining time value.
- Right tail (very high S): now the plateau is

$$\Pi_{T_1}(S) \approx K_1 - K_2 e^{-r\tau} - \Pi_0,$$

  which is typically **higher** than in (a) and often **positive** (since $K_1 > K_2$).
  Bullish diagonal: benefits if price rises; profits plateau at a **positive** level.

